ANDERSON JUNIOR COLLEGE

2018 JC 2 PRELIMINARY EXAMINATIONS

NAME:_________________________ **PDG:**____ /17

CHEMISTRY **9729/04**

Paper 4 Practical **28 August 2018**

2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials: As listed in the Confidential Instructions

READ THESE INSTRUCTIONS FIRST

Write your name, PDG and register number on all the work you hand in.
Give details of the practical shift and laboratory where appropriate, in the boxes provided.
Write in dark blue or black pen.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.
Quantitative Analysis Notes are printed on pages 22 and 23.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	
2	
3	
Total	

This document consists of **23** printed pages.

Answer **all** the questions in the spaces provided.

1 Determination of concentration of two solutions

You are provided with the following.

FA 1 is a dilute solution of sulfuric acid, H_2SO_4.
FA 2 is a dilute solution of sodium hydroxide, NaOH.
FA 3 is solid anhydrous sodium carbonate, Na_2CO_3.
Methyl orange indicator solution.

Both sodium hydroxide and sodium carbonate are bases which will neutralise sulfuric acid.

In this question, you will prepare a standard solution of sodium carbonate, **FA 4**, using solid anhydrous sodium carbonate, **FA 3**. You will then add different volumes of **FA 4** to identical samples of dilute sulfuric acid, **FA 1**. In each case, the amount of sodium carbonate solution you add will only partially neutralise the sulfuric acid. You will then complete the neutralisation of each mixture by titration with dilute sodium hydroxide, **FA 2**.

Each titration is to be performed **once only**, so great care should be taken that you do not overshoot the end–points.

Graphical analysis of your results will enable you to determine the concentrations of the sulfuric acid in **FA 1** and of the sodium hydroxide in **FA 2**.

(a) (i) Preparation of solution FA 4

1. Weigh a **dry** empty weighing bottle.
2. Reweigh the bottle with between 9.8 g and 10.2 g of **FA 3**.
3. Transfer the contents of the weighing bottle into a 250 cm^3 beaker and dissolve the solid in about 100 cm^3 of deionised water.
4. Transfer all of the solution to a 250 cm^3 graduated flask.
5. Make up the solution with deionised water and mix thoroughly. This is solution **FA 4**.
6. Reweigh the weighing bottle.

In an appropriate format in the space below, record all measurements of mass. Include the mass of **FA 3** used.

Results

[2]

(ii) Preparing titration mixtures and titrations

Remember, each titration is to be performed **once only**, so great care should be taken that you do not overshoot the end–points.

1. Pipette 25.0 cm^3 of **FA 1** into a conical flask.

2. Fill a burette with **FA 4**.

3. Add 6.00 cm^3 of **FA 4** into the same conical flask. Swirl to mix the contents.

4. Fill the second burette with **FA 2**.

5. Add 2 – 3 drops of methyl orange indicator to the conical flask.

6. Titrate the mixture in the flask with **FA 2** until an orange colour is obtained.

   Record your results in Table 1.1. Make certain that your recorded results show the precision of your working.

7. Rinse the conical flask thoroughly.

8. Repeat the procedure used for steps 1 to 7 but this time, add 18.00 cm^3 of **FA 4** to 25.0 cm^3 of **FA 1**.

9. You are to perform **two additional** titrations, **each** with a **different** volume of **FA 4** added to 25.0 cm^3 of **FA 1.**

   Select **two** other suitable volumes of **FA 4** for use in two additional experiments. Your selected volumes must be between the volumes used earlier.

   Do not use volumes of FA 4 outside the range of 6 – 18 cm^3.

   Repeat the procedure in steps 1 to 7.

   You should perform your titrations in order of **increasing** volumes of **FA 4**.

Results

Table 1.1

titration number				
volume of **FA 4** used / cm^3				

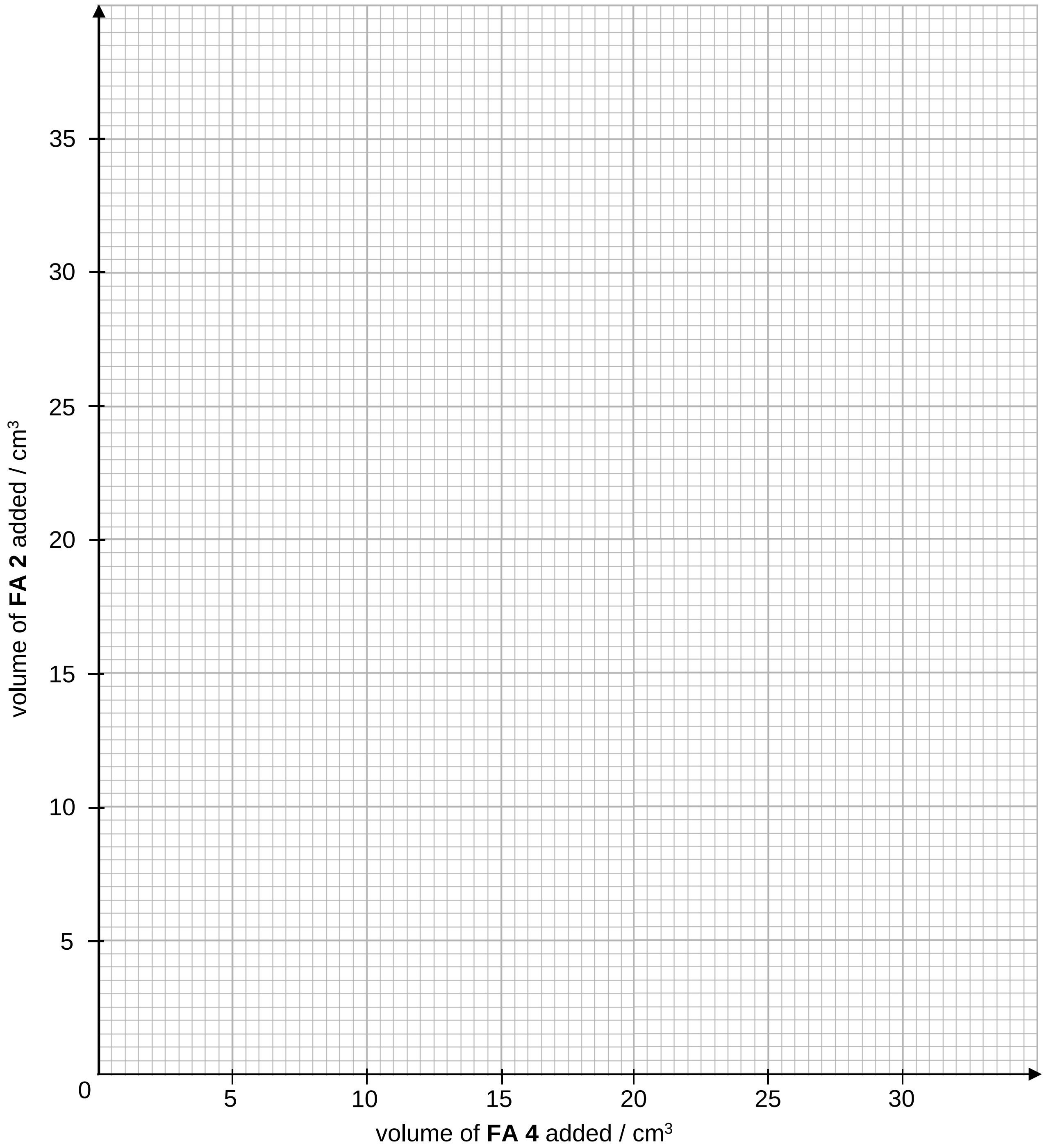
[2]

(b) Plot, on the grid below, your values for the **FA 2** titre against the volume of **FA 4** added.

The scales for the axes were chosen to allow you to determine by extrapolation

- the volume of **FA 4** required, **V_{max}(FA 4)**, to react completely with 25.0 cm^3 of **FA 1** if no **FA 2** is added;
- the volume of **FA 2** required, **V_{max}(FA 2)**, to react completely with 25.0 cm^3 of **FA 1** if no **FA 4** is added.

Draw the line of best fit, taking into account all of your plotted points. Hence, obtain values for **V_{max}(FA 4)** and **V_{max}(FA 2)**.



V_{max}(FA 4) = …………………… **V_{max}(FA 2)** = …………………… [5]

(c) Calculate the gradient of your graph line, showing clearly how you did this.

Gradient = ……………………………………….. [1]

(d) Explain, in terms of the chemistry involved, the direction of the slope of your graph.

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…………………………………………………………………………………………………… [1]

Calculations

(e) **(i)** Calculate the concentration, in mol dm^{-3}, of Na_2CO_3 in the **FA 4** solution you have prepared.

[A_r: Na, 23.0; C, 12.0; O, 16.0]

concentration of Na_2CO_3 in **FA 4** = ……………………………………….. [1]

(ii) Using appropriate data from your graph, calculate the concentration of H_2SO_4 in **FA 1**.

concentration of H_2SO_4 in **FA 1** = ……………………………………….. [2]

(iii) Using your answer to **(e)(ii)** and appropriate data from your graph, calculate the concentration of NaOH in **FA 2**.

concentration of NaOH in **FA 2** = ……………………………………….. [1]

(f) Another method to calculate the concentration of NaOH in **FA 2** is to use the expression below.

$$[NaOH] = \frac{2[Na_2CO_3]}{|gradient|}$$

Use the expression to determine the concentration of NaOH in **FA 2**.

concentration of NaOH in **FA 2** = ……………………………………….. [1]

(g) A student performs this experiment. Unknowingly, he uses a sample of **FA 3** which is slightly damp.

Suggest what effect this will have on the values of **V_{max}(FA 4)** and **V_{max}(FA 2)** he obtains. In each case, explain your answer.

Effect on **V_{max}(FA 4)** ……………………………………………………………………………

Explanation ……………………………………………………………………………………

……………………………………………………………………………………………………

……………………………………………………………………………………………………

Effect on **V_{max}(FA 2)** ……………………………………………………………………………

Explanation ……………………………………………………………………………………

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…………………………………………………………………………………………………… [2]

[Total: 18]

2 Determination of the enthalpy change of a reaction

You will follow the instructions to obtain the enthalpy changes, ΔH_1 and ΔH_2 for the following reactions.

$$NaOH(aq) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l) \qquad \Delta H_1$$

$$Na_2CO_3(aq) + 2HCl(aq) \longrightarrow 2NaCl(aq) + CO_2(g) + H_2O(l) \qquad \Delta H_2$$

You will then use the results of your experiments to calculate the enthalpy change, ΔH_3 for the following reaction.

$$2NaOH(aq) + CO_2(g) \longrightarrow Na_2CO_3(aq) + H_2O(l) \qquad \Delta H_3$$

You are provided with the following.

FA 5 is 2.00 mol dm^{-3} hydrochloric acid, HCl.
FA 6 is 1.00 mol dm^{-3} sodium hydroxide, $NaOH$.
FA 7 is 1.00 mol dm^{-3} sodium carbonate, Na_2CO_3.

(a) Reaction between FA 5 and FA 6

Record your results for this experiment in Table 2.1.

1. Using a measuring cylinder, transfer 40.0 cm^3 of **FA 5** into a Styrofoam cup. Place this cup into a second Styrofoam cup, which is placed in a 250 cm^3 beaker.
2. Stir and measure the temperature of **FA 5**.
3. Using another measuring cylinder, measure 50.0 cm^3 of **FA 6**.
4. Stir and measure the temperature of **FA 6**.
5. Slip the thermometer through the lid of the cup. Add **FA 6** from the measuring cylinder into **FA 5** in the Styrofoam cup.
6. Using the thermometer, stir the mixture continuously until it reaches its maximum temperature. Record this temperature, T_{max}.

Complete Tables 2.1 and 2.2 by calculating values for the weighted average initial temperature, T_{av}, for the two solutions, and the temperature change during the reaction, ΔT_{max}.

The value for T_{av} can be calculated by using the formula given below.

$$\boldsymbol{T}_{av} = \frac{(\text{volume } \textbf{FA 5} \text{ x initial temp } \textbf{FA 5}) + (\text{volume } \textbf{FA 6} \text{ x initial temp } \textbf{FA 6})}{\text{total volume of reaction mixture}}$$

Results

Table 2.1

Initial temperature of **FA 5**	/ ^oC	
Initial temperature of **FA 6**	/ ^oC	
Average temperature of **FA 5** and **FA 6,** T_{av}	/ ^oC	
Maximum temperature after reaction, T_{max}	/ ^oC	
Temperature change, ΔT_{max}	/ ^oC	

[1]

(b) Reaction between FA 5 and FA 7

Record your results for this experiment in Table 2.2.

1. Empty and rinse the Styrofoam cup used in **(a)**. Place this cup into a second Styrofoam cup which is placed in a 250 cm^3 beaker.
2. Using a measuring cylinder, transfer 40.0 cm^3 of **FA 5** into the Styrofoam cup.
3. Stir and measure the temperature of **FA 5**.
4. Rinse the measuring cylinder used for **FA 6** in **(a)** and use it to place 50.0 cm^3 of **FA 7**.
5. Stir and measure the temperature of **FA 7**.
6. Slip the thermometer through the lid of the cup. Add **FA 7** from the measuring cylinder into **FA 5** in the Styrofoam cup.
7. Using the thermometer, stir the mixture continuously until it reaches its maximum temperature. Record this temperature, T_{max}.

Results

Table 2.2

Initial temperature of **FA 5**	/ ^oC	
Initial temperature of **FA 7**	/ ^oC	
Average temperature of **FA 5** and **FA 7,** T_{av}	/ ^oC	
Maximum temperature after reaction, T_{max}	/ ^oC	
Temperature change, ΔT_{max}	/ ^oC	

[3]

Calculations
In the following calculations, you should assume that the

- specific heat capacity of the mixture is 4.18 J g^{-1} K^{-1},
- density of the mixture is 1.00 g cm^{-3}.

(c) Enthalpy change of reaction of aqueous sodium hydroxide and hydrochloric acid, ΔH_1

(i) By reference to the volumes used, the concentrations of the solutions and the equation for the reaction, show which of the reagents **FA 5** or **FA 6** was added in excess.

[1]

(ii) Calculate the heat change, q, for your experiment in **(a)** and hence determine a value for the enthalpy change, ΔH_1, for the reaction

$$NaOH(aq) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l)$$

q = …………………………………………..

ΔH_1 = ……………………………………… [2]

(d) **Enthalpy change of reaction of aqueous sodium carbonate and hydrochloric acid, ΔH_2**

(i) By reference to the volumes used, the concentrations of the solutions and the equation for the reaction, show which of the reagents **FA 5** or **FA 7** was added in excess.

[1]

(ii) Calculate the heat change, q, for your experiment in **(b)** and hence determine a value for the enthalpy change, ΔH_2, for the reaction

$$Na_2CO_3(aq) + 2HCl(aq) \longrightarrow 2NaCl(aq) + CO_2(g) + H_2O(l)$$

q = ……………………………………………..

ΔH_2 = ……………………………………… [2]

(e) **Enthalpy change of reaction of aqueous sodium hydroxide and carbon dioxide, ΔH_3**

Use the equations

$$NaOH(aq) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l) \qquad \Delta H_1$$

$$Na_2CO_3(aq) + 2HCl(aq) \longrightarrow 2NaCl(aq) + CO_2(g) + H_2O(l) \qquad \Delta H_2$$

and your calculated values of ΔH_1 and ΔH_2 to calculate the enthalpy change, ΔH_3, for the reaction

$$2NaOH(aq) + CO_2(g) \longrightarrow Na_2CO_3(aq) + H_2O(l) \qquad \Delta H_3$$

Show your working clearly.

ΔH_3 = ………………………………………. [4]

(f) Planning

You will plan an additional experiment whose results, when linked to those from experiments in **(a)** and **(b)**, will enable you to determine the enthalpy change, ΔH_4, for the following reaction.

$$2NaOH(aq) + CO_2(g) \longrightarrow Na_2CO_3(s) + H_2O(l) \qquad \Delta H_4$$

You may assume that you are provided with the reagents and apparatus used in **(a)** and the following additional materials

- solid sodium carbonate, Na_2CO_3,
- deionised water,
- equipment normally found in a college laboratory.

In your plan, you will
- suggest the additional enthalpy change that can be used together with results from experiments in **(a)** and **(b)** and show how they can be used to determine the enthalpy change, ΔH_4,
- give a brief outline of the procedure that you would follow and the measurements that you would take.

[A_r : Na, 23.0; C, 12.0; O, 16.0]

DO NOT CARRY OUT THE PLAN.

You may show your working in the space below.

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[Total: 17]

3 Organic Qualitative Analysis

(a) You are provided with samples of **FA 8**, **FA 9**, **FA 10** and **FA 11**. Each of which is a different one of the following.

an alcohol
an aldehyde
a carboxylic acid
a ketone

Perform the tests described in the table below, and record your observations in the spaces provided in Table 3.1. You do **not** need to perform those with observations recorded.

Use a fresh sample of each solution for each test.

Table 3.1

	test	**observations**			
		FA 8	**FA 9**	**FA 10**	**FA 11**
(i)	Place about 1 cm depth of the unknown in a test tube. To the test–tube, add about 1 cm depth of 2,4–dinitrophenylhydrazine reagent. Shake and warm the test tube in a water–bath.	no observable change			

	test	observations			
		FA 8	FA 9	FA 10	FA 11
(ii)	Place about 1 cm depth of the unknown in a test tube. To the test–tube, add a small amount of magnesium powder.		no observable change		
(iii)	Place about 1 cm depth of the unknown in a test tube. To the test–tube, add 1 cm depth of Fehling's solution. Shake and warm the test–tube in a water–bath.			no observable change	
(iv)	Place about 1 cm depth of the unknown in a test tube. To this test–tube, add about 3 cm depth of iodine solution. Then add sodium hydroxide dropwise until the iodine colour has almost discharged. Warm the test–tube in a water–bath if necessary.				no observable change

[3]

(b) **(i)** From the observations, give evidence for the type of compound present in **FA 8 – FA 11**.

	compound type	**evidence**
FA 8		
FA 9		
FA 10		
FA 11		

[4]

(ii) Identify the sample in which a copper containing product is formed in test **(a)(iii)**. State the identity of the copper containing product and hence the type of reaction taking place in that test tube.

sample: ……………………………………………………………………………………….

identity: ………………………………………………………………………………………..

type of reaction: ………………………………………………………………………….... [1]

(c) Planning

2,4–dinitrophenylhydrazine can be synthesised from benzene by a 3–step synthesis with bromobenzene and 2,4–dinitrobromobenzene as intermediates. The preparation of organic compounds usually produces a mixture of the required compound and other impurities.

The synthesis of bromobenzene is described below. When excess bromine is used in the synthesis of bromobenzene, 1,4–dibromobenzene is formed as a by–product.

Preparation and purification of bromobenzene

20.0 cm^3 of benzene, 6.0 cm^3 of bromine are mixed with an iron tack to catalyse the reaction. The mixture is warmed at 50 – 55 °C. During the reaction, acidic fumes of hydrogen bromide is evolved. After about 15 min when the reaction subsides, the reaction mixture is removed from the water–bath.

The mixture is then heated to drive off unreacted bromine and cooled before doing solvent extraction. 25 cm^3 of ether is added to dissolve the mixture before shaking the ether solution with two 5 cm^3 portions of sodium hydroxide and one 10 cm^3 portion of water, keeping only the ether layer each time and disposing the aqueous layer. A spatula full of anhydrous magnesium sulfate is added to dry the ether layer for a few minutes before it is filtered away. Fractional distillation is done and only the bromobenzene fraction is collected.

The procedure above, when excess benzene was used, achieved a yield of 75%. A student would like to investigate if increasing the proportion of bromine used, by reducing amount of benzene in the reaction mixture, can raise the yield of bromobenzene.

Some relevant data are given in Table 3.2.

Table 3.2

Compound	boiling point / °C	density / g cm^{-3}	M_r
bromine	59	3.12	159.8
benzene	80	0.88	78.0
bromobenzene	156	1.50	156.9
1,4–dibromobenzene	220	1.84	235.8

(i) Write a balanced equation for the formation of bromobenzene from benzene.

………………………………………………………………………………………………….. [1]

(ii) It is also found from literature that when the amount of bromine used is twice the amount of benzene, 1,4–dibromobenzene is formed as the major product instead.

Calculate the volume of benzene that you would use to react with 6.0 cm^3 of bromine such that the amount of bromine is twice that of benzene.

[2]

(iii) The optimum ratio of bromine to benzene can be determined by repeating the experiment several times using different initial ratios of reactants each time.

Using the information given on page 18 and your answer to **(c)(ii)**, suggest the different volumes of reactants to be used in each experiment and complete Table 3.3.

Table 3.3

experiment number	volume of benzene used / cm^3	volume of bromine used / cm^3
1	20.0	6.0
2		
3		

[1]

(iv) You are required to write a plan to determine the optimum ratio of bromine to benzene to achieve the highest possible yield of bromobenzene.

Your plan should include the following:

- give a full description of the procedure that you would use to prepare and purify bromobenzene using the three different reaction mixtures in **(c)(iii)**,
- an outline of how percentage yield (with reference to amount of benzene used) would be obtained from each of the three different reaction mixtures,
- any safety precaution that you consider important.

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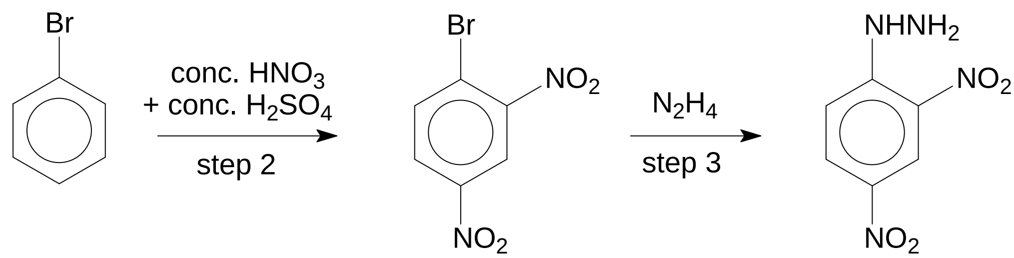
………………………………………………………………………………………….. [6]

(v) The bromobenzene obtained is then converted to 2,4–dinitrophenylhydrazine as shown in the following reaction scheme.



Name the types of reaction that occur during each of the steps.

Step 2: ………………………………………………………

Step 3: ……………………………………………………… [2]

[Total: 20]

Qualitative Analysis Notes
[ppt. = precipitate]

(a) Reactions of aqueous cations

cation	*reaction with*	
	NaOH(aq)	NH_3(aq)
aluminium, Al^{3+}(aq)	white ppt. soluble in excess	white ppt. insoluble in excess
ammonium, NH_4^+(aq)	ammonia produced on heating	–
barium, Ba^{2+}(aq)	no ppt. (if reagents are pure)	no ppt.
calcium, Ca^{2+}(aq)	white ppt. with high [Ca^{2+}(aq)]	no ppt.
chromium(III), Cr^{3+}(aq)	grey–green ppt. soluble in excess giving dark green solution	grey–green ppt. insoluble in excess
copper(II), Cu^{2+}(aq)	pale blue ppt. insoluble in excess	blue ppt. soluble in excess giving dark blue solution
iron(II), Fe^{2+}(aq)	green ppt. turning brown on contact with air insoluble in excess	green ppt. turning brown on contact with air insoluble in excess
iron(III), Fe^{3+}(aq)	red–brown ppt. insoluble in excess	red–brown ppt. insoluble in excess
magnesium, Mg^{2+}(aq)	white ppt. insoluble in excess	white ppt. insoluble in excess
manganese(II), Mn^{2+}(aq)	off–white ppt. rapidly turning brown on contact with air insoluble in excess	off–white ppt. rapidly turning brown on contact with air insoluble in excess
zinc, Zn^{2+}(aq)	white ppt. soluble in excess	white ppt. soluble in excess

(b) Reactions of anions

ion	*reaction*
carbonate, CO_3^{2-}	CO_2 liberated by dilute acids
chloride, Cl^-(aq)	gives white ppt. with Ag^+(aq) (soluble in NH_3(aq))
bromide, Br^-(aq)	gives pale cream ppt. with Ag^+(aq) (partially soluble in NH_3(aq))
iodide, I^-(aq)	gives yellow ppt. with Ag^+(aq) (insoluble in NH_3(aq))
nitrate, NO_3^-(aq)	NH_3 liberated on heating with OH^-(aq) and A*l* foil
nitrite, NO_2^-(aq)	NH_3 liberated on heating with OH^-(aq) and A*l* foil NO liberated by dilute acids (colourless NO $\rightarrow$ (pale) brown NO_2 in air)
sulfate, SO_4^{2-}(aq)	gives white ppt. with Ba^{2+}(aq) (insoluble in excess dilute strong acids)
sulfite, SO_3^{2-}(aq)	SO_2 liberated on warming with dilute acids; gives white ppt. with Ba^{2+}(aq) (soluble in dilute strong acids)

(c) Tests for gases

gas	*test and test result*
ammonia, NH_3	turns damp red litmus paper blue
carbon dioxide, CO_2	gives a white ppt. with limewater (ppt. dissolves with excess CO_2)
chlorine, Cl_2	bleaches damp litmus paper
hydrogen, H_2	"pops" with a lighted splint
oxygen, O_2	relights a glowing splint
sulfur dioxide, SO_2	turns aqueous acidified potassium manganate(VII) from purple to colourless

(d) Colour of halogens

halogen	*colour of element*	*colour in aqueous solution*	*colour in hexane*
chlorine, Cl_2	greenish yellow gas	pale yellow	pale yellow
bromine, Br_2	reddish brown gas / liquid	orange	orange–red
iodine, I_2	black solid / purple gas	brown	purple